SATYAJIT S

B.Tech. Computer Science • IIITDM Kancheepuram
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Computer science undergraduate working on robotic manipulation, control and teleoperation, and on physical AI – applying reinforcement learning and vision-language-action models to robot control.

EDUCATION

IIITDM Kancheepuram – B.Tech. Computer Science *2024 – 2028 | CGPA: 8.90*
The Lords International School, Chennai – Cambridge Board (A-Levels) *2023 – 2024 | 80.8%*

TECHNICAL SKILLS

- **Robotics & Control:** ROS 2, Nav2, MoveIt 2, Gazebo, RViz 2, MuJoCo, inverse kinematics, PID control, teleoperation, LiDAR SLAM (slam_toolbox), mapping and localization, ArUco pose estimation
- **Programming Languages:** C, C++, Python, JavaScript
- **Machine Learning & Computer Vision:** PyTorch, TensorFlow, Stable-Baselines3, OpenCV, YOLO, Scikit-learn
- **Tools:** Git, Linux, Docker

EXPERIENCE

e-Yantra Robotics Research Intern | IIT Bombay *May 2026 – Jul 2026*

- Selected for the eYSIP research programme on the strength of a top-20 national finish in the e-Yantra Robotics Competition.
- Trained and benchmarked four Vision-Language-Action (VLA) policies – ACT (Action Chunking with Transformers), SmolVLA, π_0 and GR00T – on SO-101 bimanual arms using the LeRobot framework, comparing task success across architectures.
- Collected expert demonstration datasets for imitation learning and deployed the policies on bimanual stacking, precision pick-and-place, and arranging markers and cups.

MaRS Student Rover Society (IIITDM) | Software Team Member *Jan 2025 – Present*

- Developed a Cartesian control pipeline for a custom-built 5-DOF manipulator using MoveIt 2: the operator commands the end-effector directly while inverse kinematics resolves joint angles, replacing impractical joint-by-joint teleoperation.
- Applied the arm to teleoperated panel tasks – pressing buttons, actuating switches and levers – requiring precise end-effector positioning at distance.
- Contributed to the rover software stack that placed the team **10th at the International Rover Challenge (IRC)**.

PROJECTS

Autonomous Holonomic Bot | e-Yantra Robotics Competition (IIT Bombay) *Sep 2025 – May 2026*

- Placed **top 20 of 300+ teams nationwide**, clearing six months of national software and hardware qualifying rounds.
- Built a holonomic robot with a PID-controlled omnidirectional drive for precise trajectory tracking and docking, transporting crates to drop zones.
- Integrated an ArUco marker vision pipeline for real-time pose estimation, driving automated detection and pick-and-place.

TD3-based Autonav | Reinforcement Learning Navigation *Sep 2026 – Oct 2026*

- Developed a TD3 agent for TurtleBot3 maze navigation in Gazebo (ROS 2 Humble, Stable-Baselines3) using LiDAR and relative-goal observations, with BFS geodesic reward shaping.
- Trained on a single maze and transferred to unseen layouts without retraining, reaching $\sim$**98% success** on the training map.

RESEARCH

RL-Based QA Tester for Games | ViZDoom, PyTorch, PPO, Go-Explore *Oct 2026 – Present*

- Built the environment end to end: a declarative map specification compiles to a playable UDMF Doom level plus ground-truth JSON, driving a ViZDoom simulation in which six game-agnostic oracle families detect injected defects.
- Training the RL tester now – PPO with count-based curiosity and Go-Explore archive resets, benchmarked against a task-reward-only baseline and scored on precision and recall against labelled ground truth.

CERTIFICATIONS

- **Udemy:** Self-Driving – Map & Localization | Self-Driving – Odometry and Control | Python for Data Science and ML